

# Unit 6

## (Introduction of Electromagnetism)

### Divergence

$$\text{div } \vec{A} = \nabla \cdot \vec{A} = \frac{\partial A}{\partial x} + \frac{\partial A}{\partial y} + \frac{\partial A}{\partial z}$$

$$\text{div } \vec{A} = \oint_V \vec{A} \cdot d\vec{s}$$

### Divergence Gauss theorem

$$\int_V (\nabla \cdot \vec{A}) dV = \oint_S \vec{A} \cdot d\vec{s}$$

$$(\text{div } \vec{E} = \rho / \epsilon_0)$$

### Poisson's equation

$$\boxed{\nabla^2 \phi = -\rho / \epsilon_0}$$

here  $\nabla^2 \phi = \left( \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} \right)$

### Biot-Savart law

$$\vec{H} = \frac{1}{4\pi} \oint \frac{I d\vec{L} \times \vec{R}}{R^3}$$

$$(\because \vec{B} = \mu_0 \vec{H})$$

### Curl

$$\begin{aligned} \text{Curl } \vec{A} &= \nabla \times \vec{A} \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \end{aligned}$$

$$\text{Curl } \vec{A} = \nabla \times \vec{A} = \frac{\oint_L \vec{A} \cdot d\vec{l}}{\Delta S} \hat{n}$$

### Stoke's theorem

$$\oint_L \vec{A} \cdot d\vec{l} = \int_S (\nabla \times \vec{A}) \cdot d\vec{s}$$

### Laplace equation

$$\boxed{\nabla^2 \phi = 0}$$

### Faraday's law

$$\text{emf} = -N \frac{d\phi}{dt}$$

$$\because \phi = \int \vec{B} \cdot d\vec{s}$$

$$\text{emf} = -N \int \frac{d\vec{B}}{dt} \cdot d\vec{s}$$



## Displacement current

$I_d = \frac{dQ}{dt}$  Current between two plates of capacitor.

$$\oint \vec{H} \cdot d\vec{l} = \frac{dQ}{dt} = I_d$$

## Maxwell's equation

$$(i) \quad \boxed{\oint \vec{E} \cdot d\vec{s} = \frac{Q}{\epsilon} = \frac{1}{\epsilon} \int \rho dv}$$

diff. form  $\Rightarrow \text{div } \vec{E} = \vec{\nabla} \cdot \vec{E} = \rho / \epsilon$

$$(ii) \quad \boxed{\oint \vec{B} \cdot d\vec{s} = 0}$$

$$\boxed{\vec{\nabla} \cdot \vec{B} = \text{div } B = 0}$$

$$(iii) \quad \mathcal{E} = - \frac{d\phi_B}{dt} = \oint \vec{E} \cdot d\vec{r} = - \oint \frac{d\vec{B}}{dt} \cdot d\vec{s}$$

$$\boxed{\text{Curl } \vec{E} = \vec{\nabla} \times \vec{E} = - \frac{d\vec{B}}{dt}}$$

$$(iv) \quad \boxed{\oint \vec{B} \cdot d\vec{r} = \mu_0 \Sigma (I_c + I_d)}$$

$$\oint \vec{B} \cdot d\vec{r} = \mu_0 \oint_s \left( \vec{J} + \frac{d\vec{D}}{dt} \right) \cdot d\vec{s}$$

or

$$\vec{\nabla} \times \vec{B} = \mu \left( \vec{J}_f + \frac{d\vec{D}}{dt} \right) = \left( \mu_0 \vec{J}_f + \mu \vec{E} \frac{d\vec{E}}{dt} \right)$$

$$\vec{\nabla} \times \vec{B} = \mu \vec{J}_f$$



Poynting vector =  $\frac{\text{energy}}{\text{Area} \times \text{time}} = \frac{P}{A}$

$$P = \vec{E} \times \vec{H} = \frac{\vec{E} \times \vec{B}}{\mu}$$

## Unit 5 Semiconductor & material Sci.

### Types of bond

- Ionic → High electro negativity diff.
- Covalent → electronegativity ↓ & non metallic.
- Metallic → between metal crystal
- Vander wall's

Formation energy bands ⇒ Range of energy possessed by an electron.

### Fermi Distribution function

$$f(E) = \frac{1}{1 + e^{(E-E_F)/kT}}$$

$k$  = boltzmann constant  
 $T$  = Temp

### Electrical conductivity of semiconductor.

$$\sigma = \frac{J}{E} = e(n\mu_e + p\mu_h) \quad \left| \begin{array}{l} \text{Resistivity} \\ \rho = \frac{1}{\sigma} \end{array} \right.$$

### Hall effect

$$E_H = R_H B J$$

$B$  = magnetic field

$J$  =

&

$$R_H = -\frac{1}{ne}$$

Mobility

$$\mu_e = -e R_H$$

### Unit 3   Optical Fibre

Coherence Length

$$L_c = \frac{c}{\Delta\nu} = \frac{\lambda^2}{\Delta\lambda} = \lambda Q$$

(here  $Q = \frac{\lambda}{\Delta\lambda} = \text{Quality factor}$ )

Coherence time

$$\tau_c = \frac{L_c}{c}$$

Acceptance Angle

$$\theta_{\max} = \sin^{-1} \sqrt{n_1^2 - n_2^2}$$

$n_1 = \text{Core}$

$n_2 = \text{Cladding}$

Numerical Aperture

$$NA = \sin \theta_{\max} = \sqrt{n_1^2 - n_2^2}$$

$$NA \approx n_1 \sqrt{2\Delta}$$

$$\text{here } \Delta = \frac{n_1 - n_2}{n_1}$$



## Quantum Mechanics      Unit 2

particle Density  $|\psi|^2 = \psi \psi^* = \text{probability density.}$

→ probability  $P = \boxed{\int |\psi|^2 dv = 1}$

→ Normalized function  $\psi$  when  
 $\langle \psi | \psi \rangle = \int_{-\infty}^{\infty} |\psi|^2 dx$

Time Dependent wave equation.

$$i\hbar \frac{\partial \psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x,t)}{\partial x^2} + U \psi(x,t)$$

~~for 3D~~ Hamiltonian operator (Total energy)

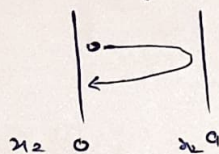
$$H = \left[ -\frac{\hbar^2}{2m} \nabla^2 + U \right]$$

→ Time independent

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{2m}{\hbar^2} [E - U(x)] \psi(x) = 0$$

Free Particle in 1D Box

general solution



limit 0 to  $a$

$$\psi(x) = A \sin Kx + B \cos Kx \quad (B=0)$$

$$P = \int_{-\infty}^{\infty} \psi \psi^* dx = 1$$

$$\psi = A$$

find  $A$  by this

## Eigen Energy

$$* \left[ E_n = \frac{n^2 h^2}{8 m a^2} \right] *$$

(for zero point energy  $n=1$ )

for 3-D

$$E = \frac{h^2}{8 m a^2} (n_x^2 + n_y^2 + n_z^2)$$

$$\left( \psi(x, y, z) = \left( \frac{2}{a} \right)^{3/2} \sin\left(\frac{n_x \pi}{a}\right) x \sin\left(\frac{n_y \pi}{b}\right) y \sin\left(\frac{n_z \pi}{c}\right) z \right)$$

→ for Degeneracy.

$$\boxed{E_a = E_b}$$



# Unit 1 Optical fibre

## wave optics

### Michelson interference

$$2d \cos \theta = (n + \frac{1}{2}) \lambda$$

(for Bright fringes)

$$2d \cos \theta = n \lambda$$

for dark fringes

by  $2d = m \lambda$

$$\lambda = \frac{2\pi}{N}$$

⊗

$$N_2 - N_1 = \frac{2\pi (\lambda_1 - \lambda_2)}{\lambda_1 \lambda_2}$$

$$\Rightarrow (\lambda_1 - \lambda_2 = \frac{\lambda_{avg}}{2\pi})^{**}$$

### fringe shift

$$* \boxed{x = (n-1)t}$$

(t = thickness)

### Newton's Ring

dark

$$2\mu t = n \lambda$$

⊗

$$D_n = \sqrt{\frac{4Rn\lambda}{\mu}}$$

bright

$$2\mu t = (n + \frac{1}{2}) \lambda$$

$$D_n = \sqrt{\frac{2(2n+1)\lambda R}{\mu}}$$

fringe width

$$\begin{cases} \beta = r_{n+1} - r_n \\ \beta = \frac{1}{2} [D_{n+1} - D_n] \end{cases}$$

### wavelength by Newton's Ring

$$\boxed{\lambda = \frac{D_{n+1}^2 - D_n^2}{4nR}}$$

n = order .

~~β~~



## Diffraction

$$\text{phase diff} = \frac{2\pi}{\lambda} e \sin \theta$$

$$\text{path diff} = e \sin \theta = \Delta x = \delta$$

$$* \boxed{I = I_0 \left( \frac{\sin^2 \alpha}{\alpha^2} \right)}$$

Intensity

$$\text{here } \alpha = \delta/2 = \frac{e \sin \theta}{2}$$

\*\*

$$\boxed{\text{Resolving Power} = \lambda / d\lambda = \frac{1}{\text{Resolution limit}}}$$

$$a + b \sin \theta_n = n\lambda \quad \text{--- i for 1}$$

$$(a+b) \sin(\theta_n + d\theta_n) = n(\lambda + d\lambda) \quad \text{--- ii for 2}$$

Resolving power

$$** \boxed{\frac{\lambda}{d\lambda} = nN}$$

Here  $N$  = number of lines per inch.

$$n = \frac{(a+b) \sin \theta}{\lambda}$$

$$\boxed{w = N(a+b)} \quad \text{grating width.}$$

\*\*

$$\frac{\lambda}{d\lambda} = N \frac{(a+b) \sin \theta}{\lambda} = \frac{w \sin \theta}{\lambda}$$

## Bragg's Condition

$$\sin \theta \leq 1$$

$$\frac{\lambda}{2d} \leq 1 \quad \Rightarrow \quad \boxed{\lambda \leq 2d}$$

Constructive interference b/w diffracted rays.